Example 2.8

If the area of the rhombus is 60 $sq.\ cm$ and one of the diagonals is 8 cm, find the length of the other diagonal.

Solution

Given, the length of one diagonal (d_1) = 8 cm

Let, the length of the other diagonal be $d_2\ cm$

$$\text{Area of the rhombus} = 60\ sq.\ cm \quad \text{(given)}$$

$$\frac{1}{2} \times (d_1 \times d_2) = 60$$

$$\frac{1}{2} \times (8 \times d_2) = 60$$

$$8 \times d_2 = 60 \times 2$$

$$d_2 = \frac{120}{8}$$

$$= 15$$

Therefore, length of the other diagonal is 15 cm.

Example 2.9

The floor of an office building consists of 200 rhombus shaped tiles and each of its length of the diagonals are 40 cm and 25 cm. Find the total cost of polishing the floor at ₹ 45 per $sq.m.$

Solution

Given, the length of the diagonals of a rhombus shaped tile are 40 cm and 25 cm

$$\text{The area of one tile} = \frac{1}{2} \times (d_1 \times d_2)\ sq.\ units$$

$$= \frac{1}{2} \times 40 \times 25$$

$$= 500\ sq.\ cm$$

$$\text{Therefore, the area of 200 such tiles} = 200 \times 500$$

$$= 100000\ sq.\ cm$$

$$= \frac{100000}{10000} (1\ sq.\ m = 10000\ sq.\ cm)$$

$$= 10\ sq.\ m$$

Therefore, the cost of polishing 200 such tiles at the rate of ₹ 45 per $sq.\ m$

= 10 x 45 = ₹ 450.



DO YOU KNOW?

In the terminology of railways, "Diamond Crossing" refers to the point where two railway lines cross, forming the shape of rhombus at the crossing point. The most famous diamond crossing is at Nagpur, where lines from the North, South, East, and Western railways meet.

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